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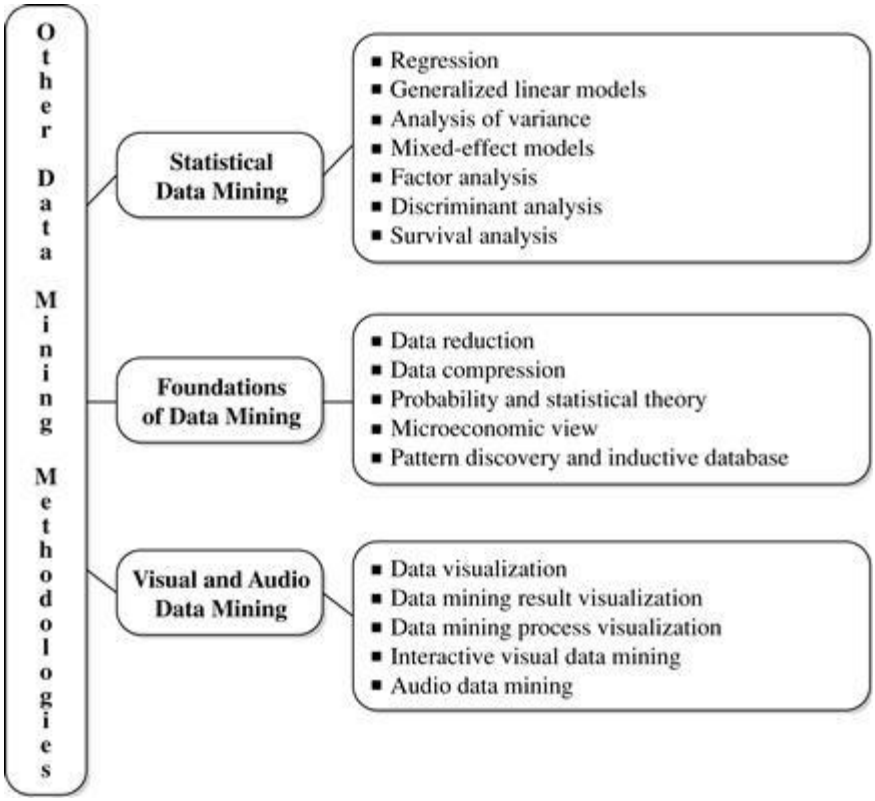
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UNIT 5

OTHER METHODOLOGIES OF DATA MINING

Due to the broad scope of data mining and the large variety of data mining methodologies, not all methodologies of data mining can be thoroughly covered in this book. In this section, we briefly discuss several interesting methodologies that were not fully addressed.



Statistical Methods in Data Mining

Data mining refers to extracting or mining knowledge from large amounts of data. In other words, data mining is the science, art, and technology of discovering large and complex bodies of data in order to discover useful patterns. Theoreticians and practitioners are continually seeking improved techniques to make the process more efficient, cost-effective, and accurate. Any situation can be analyzed in two ways in data mining:

- **Statistical Analysis:** In statistics, data is collected, analyzed, explored, and presented to identify patterns and trends. Alternatively, it is referred to as quantitative analysis.
- **Non-statistical Analysis:** This analysis provides generalized information and includes sound, still images, and moving images.

In statistics, there are two main categories:

- **Descriptive Statistics:** The purpose of descriptive statistics is to organize data and identify the main characteristics of that data. Graphs or numbers summarize the data. Average, Mode, SD(Standard Deviation), and Correlation are some of the commonly used descriptive statistical methods.

- **Inferential Statistics:** The process of drawing conclusions based on probability theory and generalizing the data. By analyzing sample statistics, you can infer parameters about populations and make models of relationships within data.

There are various statistical terms that one should be aware of while dealing with statistics. Some of these are:

- Population
- Sample
- Variable
- Quantitative Variable
- Qualitative Variable
- Discrete Variable
- Continuous Variable

Now, let's start discussing statistical methods. This is the analysis of raw data using mathematical formulas, models, and techniques. Through the use of statistical methods, information is extracted from research data, and different ways are available to judge the robustness of research outputs

Foundations of Data Mining

A Position Paper Dr. Bhavani Thuraisingham The MITRE Corporation (at present with the National Science Foundation) Data Mining is the process of posing queries to large amounts of data sources and extracting patterns and trends using statistical and machine learning techniques. It integrates various technologies including database management, statistics and machine learning. Data mining has applications in numerous disciplines including medical, financial, defense and intelligence. Data mining tasks include classification, clustering, making associations and anomaly detection. For example, data mining can extract various associations between people, places or words. During recent years there have been many developments in data mining. Various data mining techniques have been developed. These include techniques for extracting associations, neural networks, inductive logic programming, decision trees, fuzzy logic and rough sets. Furthermore, data mining has gone beyond mining relational databases to mining text and multimedia data. Also, data mining is being applied to areas such as information security and intrusion detection. While thee have been many practical developments, we still have major challenges. One of the most important challenges is scalability. For example, the data mining techniques often seem to work on small quantities of data. But do these techniques work for say petabyte sized databases? If data mining is to be useful we need to mine very large databases. Therefore, it is critical that we need to understand the limitations of the data mining algorithms. To understand the limitations, we need to study the foundations of data mining. We need to explore the time and space complexity of the algorithms. For example, can these algorithms be completed in polynomial time? Are there any undecidable problems? If the problems are decidable what is the complexity of the problems? To date little research has been carried out on the foundations of data mining. There are techniques such as inductive logic programming and rough sets that have underpinnings in logic and mathematics. One needs to explore these techniques for data mining and examine the computational complexity aspects. We also need to understand the complexity of the various search algorithms being used for market basket analysis. We are at

a time where data mining applications are exploding. Data mining is becoming one of the fastest growing fields in computer science. We are also at a time where we have a much better understanding of the problem. Therefore, the time is right to start active research on the foundations of data mining. But we also have to be cautious. That is, we need to carry out incremental research. In some cases, we may not be able to explore the foundations. Therefore, we have to be very selective. We need to decide which of the techniques are to be examined and then carry out theoretical research step by step

Visual and Audio Data Mining

Data mining is a process that interacts with a massive set of data. In this perspective, it unravels interesting patterns from unknown data structured. The same may apply to **audio and video data mining** as well. Today, users can access a large volume of multimedia data generated from information technology and the easy availability of multimedia systems. Hence, the amount of audio and video data available today grows exponentially. Video falls under the multimedia category that contains various data comprising text, image, visual, audio, and meta-data.

Audio-video mining holds a major place in different applications across security and surveillance, medicine discovery, education, entertainment, and sports. The key objective of video data mining is to extract data from video sources and discover and define patterns and dynamics.

What is Visual Data Mining?

Visual data mining uses data and knowledge visualization methods to find implicit and beneficial knowledge from huge data sets. The eyes and brain manage the human visual system, the latter of which can be thought of as a dynamic, largely parallel processing and reasoning engine, including a huge knowledge base.

Visual data mining essentially combines the power of these components, making it a highly attractive and effective tool for the comprehension of data distributions, patterns, clusters, and outliers in data.

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Visual data mining can be considered as an association of two disciplines such as data visualization and data mining. It can also associate with computer graphics, multimedia systems, human-computer interaction, pattern identification, and high-performance computing. In general, data visualization and data mining can be integrated into the following ways:



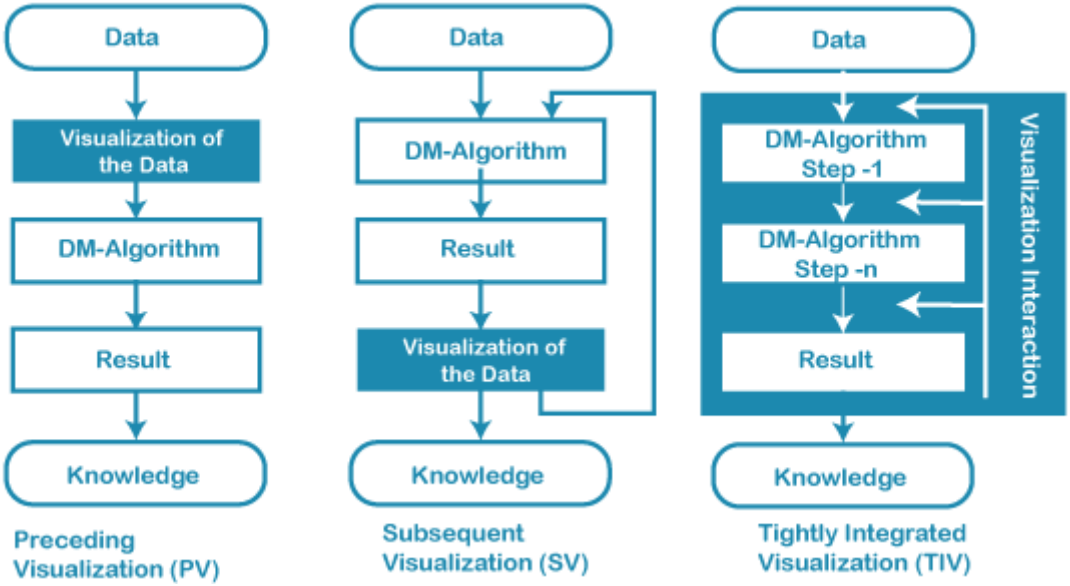
1. **Data visualization:** Data in a database or data warehouse can be viewed at different granularity or abstraction levels or as different combinations of attributes or dimensions. Data can be presented in various visual forms, including boxplots, 3-D cubes, data distribution charts, curves, surfaces, link graphs, etc. The visual display can help give users a clear impression and overview of the data characteristics in a large data set.
2. **Data mining result visualization:** Visualization of data mining results is the presentation of the results or knowledge obtained from data mining in visual forms. Such forms may include scatter plots and boxplots, decision trees, association rules, clusters, outliers, generalized rules, etc.
3. **Data mining process visualization:** This type of visualization presents the various processes of data mining in visual forms so that users can see how the data are extracted and from which database or data warehouse they are extracted, as well as how the selected data are cleaned, integrated, preprocessed, and mined. Moreover, it may also show which method is selected for data mining, where the results are stored, and how they may be viewed.
4. **Interactive visual data mining:** In interactive visual data mining, visualization tools can be used in the data mining process to help users make smart data mining decisions. For example, the data distribution in a set of attributes can be displayed using colored sectors (where a circle represents the whole space). This display helps users determine which sector should first be selected for classification and where a good split point for this sector may be.

Visual Data Mining Techniques

For data mining to be effective, it is important to include humans in the data exploration process and combine the flexibility, creativity, and general knowledge of the human with the enormous storage capacity and the computational power of today's computers. First, the data analyst typically specifies some parameters to restrict the search space; an algorithm performs data mining automatically. Finally, the patterns found by the automatic data-mining algorithm are presented to the data analyst on the screen. Since an automatic data-mining algorithm generates many patterns in textual form, it is almost impossible for humans to interpret and evaluate the pattern in detail and extract interesting knowledge and general characteristics.

Visual data mining aims to integrate humans in the data-mining process and apply human perceptual abilities to analyze large datasets available in today's computer systems. Presenting data in an interactive, graphical form often fosters new insights, encouraging the formation and validation of new hypotheses to the end of better problem-solving and gaining deeper domain knowledge.

Visual data exploration usually follows a three-step process: ***Overview, zoom*** *and* ***filter***, and then ***details-on-demand***. First, the data analyst needs to get an overview of the data. In the overview, the data analyst identifies interesting patterns or groups in the data and focuses on one or more of them. For analyzing the patterns, the data analyst needs to drill down and access the details of the data. Visualization technology may be used for all three steps of the data exploration process.



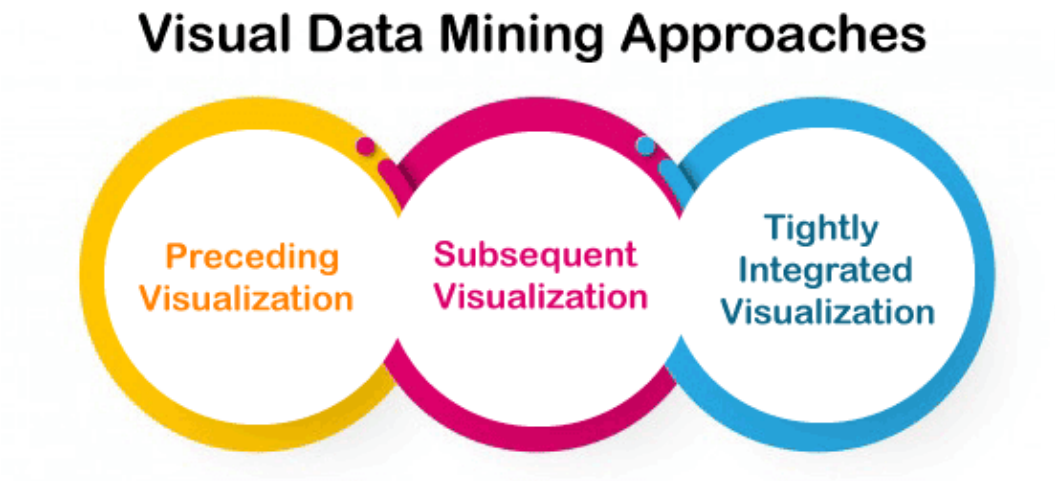
Visualization techniques are useful for showing an overview of the data, allowing the data analyst to identify interesting subsets. This step is important to keep the overview visualization while focusing on the subset using another visualization technique. This can be done by dedicating a larger display percentage to the interesting subsets while decreasing screen utilization for uninteresting data. To further explore the interesting subsets, the data analyst needs a drill-down capability to observe the details about the data.

Note that visualization technology provides the base visualization techniques for all three steps and bridges the gaps between the steps. Visual data mining can be seen as a hypothesis-generation process; the visualizations of the data allow the data analyst to gain insight into the data and come up with new hypotheses.

Verifying the hypotheses can also be done via data visualization, but it may also be accomplished by automatic techniques from statistics, pattern recognition, or machine learning. As a result, visual data mining usually allows faster data exploration and often provides better results, especially in cases where automatic data-mining algorithms fail. In addition, visual data exploration techniques provide a much higher degree of user satisfaction and confidence in the exploration findings. This fact leads to high demand for visual exploration techniques and makes them indispensable with automatic exploration techniques.

Visual Data Mining Approaches

Visual data mining is based on an automatic part, the data-mining algorithm, and an interactive part, the visualization technique. There are three common approaches to integrating humans in the data exploration process to realize different approaches to visual data mining.



- **Preceding Visualization (PV):** Data is visualized in some visual form before running a data-mining algorithm. By interaction with the raw data, the data analyst has full control over the analysis in the search space. Interesting patterns are discovered by exploring the data.
- **Subsequent Visualization (SV):** An automatic data-mining algorithm performs the task by extracting patterns from a given dataset. These patterns are visualized to make them interpretable for the data analyst. Subsequent visualizations enable the data analyst to specify feedback. Based on the visualization, the data analyst may want to return to the data-mining algorithm and use different input parameters to obtain better results.
- **Tightly Integrated Visualization (TIV):** An automatic data-mining algorithm analyzes the data but does not produces the final results. A visualization technique is used to present the intermediate results of the data exploration process. Then, the data analyst identifies the interesting patterns in visualizing the intermediate results based on his domain knowledge. Combining some automatic data-mining algorithms and visualization techniques enables specified user feedback for the next data-mining run. A motivation of this approach is to achieve independence of the data-mining algorithms from the application. A given automatic data-mining algorithm can be very useful in one domain but may have drawbacks in some other domain. Since there is no automatic data-mining algorithm suitable for all application domains, tightly integrated visualization leads to a better understanding of the data and the extracted patterns.

Advantages

In addition to the direct involvement of the human, here are the following main advantages of visual data exploration over automatic data-mining techniques, such as:

- Visual data exploration can easily deal with highly non-homogeneous and noisy data.
- Visual data exploration is intuitive and requires no understanding of complex mathematical or statistical algorithms or parameters.
- Visualization can provide a qualitative overview of the data, allowing data phenomena to be isolated for further quantitative analysis.
- Visual data-mining techniques have proven to be of high value in exploratory data analysis and have a high potential for exploring large databases. Visual data exploration is especially useful when little is known about the data and vague exploration goals. Since the data analyst is directly involved in the exploration process, shifting and adjusting the exploration goals is automatically done if necessary.

What is Audio Data Mining?

Audio data mining uses audio signals to indicate data patterns or the features of data mining results. Although visual data mining may disclose interesting patterns using graphical displays, it requires users to concentrate on watching patterns and identifying interesting or novel features within them. This can sometimes be quite tiresome.

If patterns can be transformed into sound and music, we can listen to pitches, rhythm, tune, and melody instead of watching pictures to identify anything interesting or unusual. This may relieve some of the burdens of visual concentration and be more relaxing than visual mining. Therefore, audio data mining is an interesting complement to visual mining.

Applications of Audio and Visual Mining

Here, we will discuss different use cases of audio and video data mining in businesses, such as:

1. Traffic control management

Whether traffic control or traffic violation monitoring, video data mining or analytics can be used to reach better decision making regarding traffic management, the live feed is extracted from the cameras installed upon the traffic posts. This live input is added to the database system to process it. As per the available data extracted from the video, the traffic control system will control the devices attached to the systems. If it detects anything suspicious on the road, such as the speeding driver or miscreant activity, the system alerts the system administrator via the analytics dashboard. Other progress reports on the road are also monitored and dispatched via a separate database.

2. Vehicles monitoring procedures

The video processing via video data mining can be leveraged to control various public vehicles or transport at different busy locales. AI-based video processing is one of the useful tools to monitor vehicles.

The vehicle traffic department can use a video-based vehicle monitoring system. This is used to monitor the traffic flow on highways to determine the exact speed and travel time of transports and accurate toll values. The CCTV cameras provide images of the moving transports, which are analyzed with a video image processor. The vehicles detected by the image sequencing tool are external communication output.

3. Enhanced Security with Live Video Streaming

Enhancing the security and safety of people is a primary objective of surveillance. The main objective is to track behavior, activities, and other data to ensure safety. A video processing system embedded with surveillance is important for monitoring and security.

Provided an access control, it works in line with video processing surveillance and provides useful information about a person entering the premises through a live video stream feed. Using AI-based facial recognition technology, it assigns unique identifiers to persons to enable them to enter any areas of the premises with an authorization level access.

4. Health Status Monitoring

Surveillances embedded with CCTV video streams can help detect the health status of patients in hospitals as well. A camera that captures video streams of an infant is ideal for detecting the infant's respiratory troubles using an Eulerian video magnification and optical flow algorithms. The extracted information from the camera can be used later to offer better treatment options for the ailments.

5. Customer Demographic Data on Hand

Speech recognition technology is widely used to extract data for discovering customer demographic information. The technology leverages an extensive data-mining process on your audio/video resources and identifies a different range of information about a customer. It involves gender, age, emotion and sentiment, language, and many more.

6. Automated Transcription of Audio/Video Data

Small businesses can benefit from audio/video data mining that helps transcribe different unstructured data extracted from video and audio. This converts accurate texts from audio/video sources, enabling businesses to find relevant information to increase productivity and efficiency. At the same time, the whole process helps you automate many activities such as complaint analysis, demographic analysis, legal compliance, and more.

7. Understanding Customer opinions accurately.

Your customer opinion matters most as they determine success or failure for your business. Using an AI-based data mining process involving speech recognition can help you interpret your customers' exact and accurate meaning about your product and service. In most cases, it

performs extensive data mining on the recorded calls. This gives you more scopes to establish a better strategy for customer support and a unique customer experience.

WEB MINING

Web Mining is the process of Data Mining techniques to automatically discover and extract information from Web documents and services. The main purpose of web mining is discovering useful information from the World-Wide Web and its usage patterns.

Applications of Web Mining:

Web mining is the process of discovering patterns, structures, and relationships in web data. It involves using data mining techniques to analyze web data and extract valuable insights. The applications of web mining are wide-ranging and include:

Personalized marketing:

Web mining can be used to analyze customer behavior on websites and social media platforms. This information can be used to create personalized marketing campaigns that target customers based on their interests and preferences.

E-commerce

Web mining can be used to analyze customer behavior on e-commerce websites. This information can be used to improve the user experience and increase sales by recommending products based on customer preferences.

Search engine optimization:

Web mining can be used to analyze search engine queries and search engine results pages (SERPs). This information can be used to improve the visibility of websites in search engine results and increase traffic to the website.

Fraud detection:

Web mining can be used to detect fraudulent activity on websites. This information can be used to prevent financial fraud, identity theft, and other types of online fraud.

Sentiment analysis:

Web mining can be used to analyze social media data and extract sentiment from posts, comments, and reviews. This information can be used to understand customer sentiment towards products and services and make informed business decisions.

Web content analysis:

Web mining can be used to analyze web content and extract valuable information such as keywords, topics, and themes. This information can be used to improve the relevance of web content and optimize search engine rankings.

Customer service:

Web mining can be used to analyze customer service interactions on websites and social media platforms. This information can be used to improve the quality of customer service and identify areas for improvement.

Healthcare:

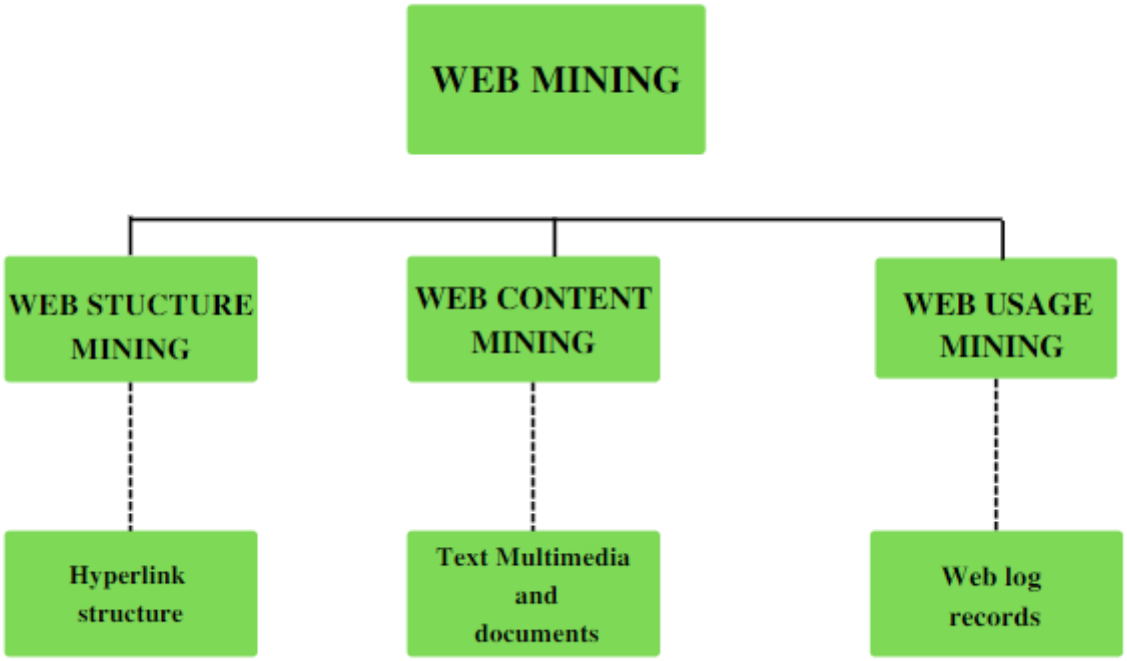
Web mining can be used to analyze health-related websites and extract valuable information about diseases, treatments, and medications. This information can be used to improve the quality of healthcare and inform medical research.

Process of Web Mining:



Web Mining Process

Web mining can be broadly divided into three different types of techniques of mining: Web Content Mining, Web Structure Mining, and Web Usage Mining. These are explained as following below.



Categories of Web Mining

1. **Web Content Mining:** Web content mining is the application of extracting useful information from the content of the web documents. Web content consist of several types of data – text, image, audio, video etc. Content data is the group of facts that a web page is designed. It can provide effective and interesting patterns about user needs. Text documents are related to text mining, machine learning and natural language processing. This mining is also known as text mining. This type of mining performs scanning and mining of the text, images and groups of web pages according to the content of the input.
2. **Web Structure Mining:** Web structure mining is the application of discovering structure information from the web. The structure of the web graph consists of web pages as nodes, and hyperlinks as edges connecting related pages. Structure mining basically shows the structured summary of a particular website. It identifies relationship between web pages linked by information or direct link connection. To determine the connection between two commercial websites, Web structure mining can be very useful.
3. **Web Usage Mining:** Web usage mining is the application of identifying or discovering interesting usage patterns from large data sets. And these patterns enable you to

understand the user behaviors or something like that. In web usage mining, user access data on the web and collect data in form of logs. So, Web usage mining is also called log mining.

Comparison Between Data mining and Web mining:

Points	**Data Mining**	**Web Mining**
Definition	Data Mining is the process that attempts to discover pattern and hidden knowledge in large data sets in any system.	Web Mining is the process of data mining techniques to automatically discover and extract information from web documents.
Application	Data Mining is very useful for web page analysis.	Web Mining is very useful for a particular website and e-service.
Target Users	Data scientist and data engineers.	Data scientists along with data analysts.
Access	Data Mining access data privately.	Web Mining access data publicly.
Structure	In Data Mining get the information from explicit structure.	In Web Mining get the information from structured, unstructured and semi-structured web pages.
Problem Type	Clustering, classification, regression, prediction, optimization and control.	Web content mining, Web structure mining.
Tools	It includes tools like machine learning algorithms.	Special tools for web mining are Scrapy, PageRank and Apache logs.
Skills	It includes approaches for data cleansing, machine learning algorithms. Statistics and probability.	It includes application level knowledge, data engineering with mathematical modules like statistics and probability.

What is the Temporal Data Mining?

Temporal data mining defines the process of extraction of non-trivial, implicit, and potentially essential data from large sets of temporal data. Temporal data are a series of primary data types, generally numerical values, and it deals with gathering beneficial knowledge from temporal data.

The objective of temporal data mining is to find temporal patterns, unexpected trends, or several hidden relations in the higher sequential data, which is composed of a sequence of nominal symbols from the alphabet referred to as a temporal sequence and a sequence of continuous real-valued components called a time series, by utilizing a set of approaches from machine learning, statistics, and database technologies.

Temporal data mining is composed of three major works such as the description of temporal data, representation of similarity measures, and mining services.

Temporal Data Mining includes processing time series, generally sequences of data, which compute values of the same attribute at a sequence of multiple time points. Pattern matching using such information, where it is searching for specific patterns of interest, has attracted considerable interest in current years.

Temporal Data Mining can include the exploitation of efficient techniques of data storage, quick processing, and quick retrieval methods that have been advanced for temporal databases.

Temporal data mining is an individual phase in the process of knowledge discovery in temporal databases that calculate temporal patterns from or fit models too, temporal data is a temporal data mining algorithm.

Temporal data mining is concerned with the analysis of temporal data and for discovering temporal patterns and consistencies in sets of temporal information. It also allows the possibility of computer-driven, automatic exploration of the data. There are various tasks in temporal mining which are as follows −

- Data characterization and comparison
- Clustering analysis
- Classification
- Association rules
- Pattern analysis
- Prediction and trend analysis

Temporal data mining has led to a new way of interacting with a temporal database and specifying queries at a much more abstract level than say, temporal structured query language permits. It also facilities data exploration for problems that are due to multiple and multi-dimensionality.

The basic goal of temporal classification is to predict temporally related fields in a temporal database based on other fields. The problem, in general, is cast as deciding the general value of the temporal variable being predicted given the different fields, the training data in which the target variable is given for each observation, and a set of assumptions representing one's prior knowledge of the problem. Temporal classification techniques are associated with the complex problem of density estimation.

What is Spatial Data Mining?

A spatial database saves a huge amount of space-related data, including maps, preprocessed remote sensing or medical imaging records, and VLSI chip design data. Spatial databases have several features that distinguish them from relational databases. They carry topological and/or distance information, usually organized by sophisticated, multidimensional spatial indexing structures that are accessed by spatial data access methods and often require spatial reasoning, geometric computation, and spatial knowledge representation techniques.

Spatial data mining refers to the extraction of knowledge, spatial relationships, or other interesting patterns not explicitly stored in spatial databases. Such mining demands the unification of data mining with spatial database technologies. It can be used for learning spatial records, discovering spatial relationships and relationships among spatial and nonspatial records, constructing spatial knowledge bases, reorganizing spatial databases, and optimizing spatial queries.

It is expected to have broad applications in geographic data systems, marketing, remote sensing, image database exploration, medical imaging, navigation, traffic control, environmental studies, and many other areas where spatial data are used.

A central challenge to spatial data mining is the exploration of efficient spatial data mining techniques because of the large amount of spatial data and the difficulty of spatial data types and spatial access methods. Statistical spatial data analysis has been a popular approach to analyzing spatial data and exploring geographic information.

The term geostatistics is often associated with continuous geographic space, whereas the term spatial statistics is often associated with discrete space. In a statistical model that manages non-spatial records, one generally considers statistical independence among different areas of data.

There is no such separation among spatially distributed records because, actually spatial objects are interrelated, or more exactly spatially co-located, in the sense that the closer the two objects are placed, the more likely they send the same properties. For example, natural resources, climate, temperature, and economic situations are likely to be similar in geographically closely located regions.

Such a property of close interdependency across nearby space leads to the notion of spatial autocorrelation. Based on this notion, spatial statistical modeling methods have been developed with success. Spatial data mining will create spatial statistical analysis methods

and extend them for large amounts of spatial data, with more emphasis on effectiveness, scalability, cooperation with database and data warehouse systems, enhanced user interaction, and the discovery of new kinds of knowledge.

APPLICATIONS OF DATA MINING

Data mining is the process of discovering meaningful new correlations, patterns, and trends by sifting through large amounts of data stored in repositories, using pattern recognition technologies as well as statistical and mathematical techniques.

It is the analysis of observational datasets to discover unsuspected relationships and to summarize the records in novel techniques that are both understandable and beneficial to the data owner. There are various applications of data mining which are as follows −

Data warehouses and data preprocessing − Data warehouses are essential for information exchange and data mining. In the area of geospatial data, but, no true geospatial data warehouse exists now.

It can be creating such a warehouse requires finding means for resolving geographic and temporal data incompatibilities, such as reconciling semantics, referencing systems, geometry, accuracy, and precision.

For mathematical software in general, methods are needed for integrating information from heterogeneous sources (including data covering different time periods) and for identifying activities. For climate and ecosystem data, for instance (which are spatial and temporal), the problem is that there are too many events in the spatial domain and too few in the temporal domain.

Mining complex data types − Scientific data sets are heterogeneous in nature, generally contain semi-structured and unstructured data, including multimedia data and geo-referenced stream data. Robust methods are needed for handling spatiotemporal data, related concept hierarchies, and complex geographic relationships (e.g., non-Euclidian distances).

Graph-based mining − It is often difficult or impossible to model several physical phenomena and processes due to the limitations of existing modeling approaches. Alternatively, labeled graphs can be used to conquer some of the spatial, topological, geometric, and other relational characteristics present in numerical data sets.

In graph modeling, every data to be mined is described by a vertex in a graph, and edges between vertices describe relationships between objects. For example, graphs can be used to model chemical structures and data generated by numerical simulations, such as fluid-flow simulations.

The success of graph modeling, however, depends on improvements in the scalability and efficiency of many classical data mining tasks, such as classification, frequent pattern mining, and clustering.

Visualization tools and domain-specific knowledge − High-level graphical user interfaces and visualization tools are needed for numerical data mining systems. These must be unified with existing domain-specific data systems and database systems to guide researchers and

general users in searching for designs, interpreting and visualizing discovered designs, and utilizing discovered knowledge in their decision making.

UBIQUITOUS AND INVISIBLE DATA MINING

Data Analytics is one of the most emerging technologies in the present-day world. With the increase in the demand for portable and remote devices like mobile phones and personal digital assistants (PDAs), the need for extracting data from these devices for analysis plays a crucial role in order to perform data analysis. Therefore, accessing data from a remote device is much needed.

Ubiquitous Data Mining (UDM) is a process of analyzing data performing concrete mining and examination of distributed and heterogeneous systems like mobile and embedded devices.

UDM is used for mining data from mobile environments like cell phones and sensors, which are constrained by limited computational resources and varying networks. It supports time-critical and real-time data needs. It is also used for intelligent analysis. Using UDM we can extract hidden classifiers and clusters.

The architecture of Ubiquitous technologies

Ubiquitous Data Mining involves the collection and storage of data, processing data, and dissemination of the result that is analyzed. In order to achieve this, we make use of 6 parts, which make up the architecture of the ubiquitous system.

Sl. No	Parts	Function
1	Devices	The component that is used for storing and processing the data. Eg: Personal Computers, Super Computers
2	Communication	The mode in which devices communicate with each other. Eg: Internet, via Centralized System
3	Users	The user who would interact with the system. It can have a Single User or can have Multiple Users.
4	Control	The component that administers all the above-mentioned parts. It can either be controlled by a single administrator or by multiple administrators.
5	Data	The type of data that is stored and processed by the system. It also gives the implication to its dynamics and organizations. The data can be of the following types: • Static data • Dynamic Data

Sl. No	Parts	Function
6	Infrastructure	The infrastructure that the system employs for data discovery. Eg: Web, Database

Application of UDM:

1. **Traffic Safety:** Abnormal traffic can be detected using sensors and the data can be stored and analyzed in the system. This analyzed data can then be used to detect traffic mishaps on a real-time basis using sensors. Thus, traffic and road safety is monitored.
2. **Health Care:** The sensors can be used in creating smart homes for elderly persons and people who are in need of continuous medical attention. The sensors can be used for notifying the immediate medical need and would be collected. The data so collected can help to provide them with timely medical aids.
3. **Crisis and Calamities Management:** The previous crisis and calamities data are collected and stored and analyzed. During times of crisis, sensors can detect the crisis and the data is sent to the controllers. Results are predicted before the effect can get disastrous. Thus, helps in Crisis Management.

Invisible Data Mining

Data mining is present in all the major aspects of our life. This requires effective mining of data mining without disclosing private information via extraction of data to outsiders.

Invisible Data Mining is a process of data mining where the functionalities are performed invisibly.

Applications of Invisible Data Mining

1. Search Engines
2. Intelligent Database System
3. e-mail managers

Ubiquitous Data Mining:

Advantages:

Improved decision-making: Ubiquitous data mining can provide valuable insights into customer behavior, market trends, and other important factors, enabling organizations to make more informed decisions.

Personalized services: Ubiquitous data mining can enable personalized services, such as customized product recommendations or personalized healthcare treatments, that can improve the overall customer experience.

Improved efficiency: Ubiquitous data mining can help streamline business processes, enabling organizations to operate more efficiently.

Disadvantages:

Privacy concerns: Ubiquitous data mining can raise privacy concerns, as it involves collecting and analyzing data about individuals without their explicit consent. This can result in the disclosure of sensitive information, which can have negative consequences for individuals.
Security risks: The data used in ubiquitous data mining can be subject to security risks, such as unauthorized access or hacking, which can result in the exposure of sensitive information.
Bias: Ubiquitous data mining algorithms can be biased, which can lead to discriminatory outcomes or reinforce existing biases.

Invisible Data Mining:

Advantages:

Less intrusive: Invisible data mining can be less intrusive than other forms of data mining, as individuals may be less aware that their data is being collected and analyzed.
Improved decision-making: Invisible data mining can provide valuable insights into customer behavior, market trends, and other important factors, enabling organizations to make more informed decisions.
Personalized services: Invisible data mining can enable personalized services, such as customized product recommendations or personalized healthcare treatments, that can improve the overall customer experience.

Disadvantages:

Lack of transparency: Invisible data mining can lack transparency, as individuals may be unaware that their data is being collected and analyzed.
Privacy concerns: Invisible data mining can raise privacy concerns, as it involves collecting and analyzing data about individuals without their explicit consent.
Security risks: The data used in invisible data mining can be subject to security risks, such as unauthorized access or hacking, which can result in the exposure of sensitive information.
Don't miss your chance to ride the wave of the data revolution! Every industry is scaling new heights by tapping into the power of data. Sharpen your skills, become a part of the hottest trend in the 21st century

PRIVACY, SECURITY AND SOCIAL IMPACTS OF DATA MINING

Data Mining is to intelligently discover useful information from large amounts of data to solve real-life problems. It is a combination of two words: data and mining. Data is a collection of instances, and mining is designed to filter useful information. Data mining, called knowledge discovery in databases (KDD), is responsible for analyzing data from different perspectives and classifying them. There are many data mining techniques used to transform raw data into useful data. It has various applications such as detecting anomalous behavior, detecting fraud and abuse, terrorist activities, and investigating crimes through lie detection. Data mining can offer many benefits by improving customer service and satisfaction, and lifestyle, in general. Data mining is present in many aspects of our daily lives, whether we realize it or not. It affects how we shop, work, what we search.

Importance of data mining:

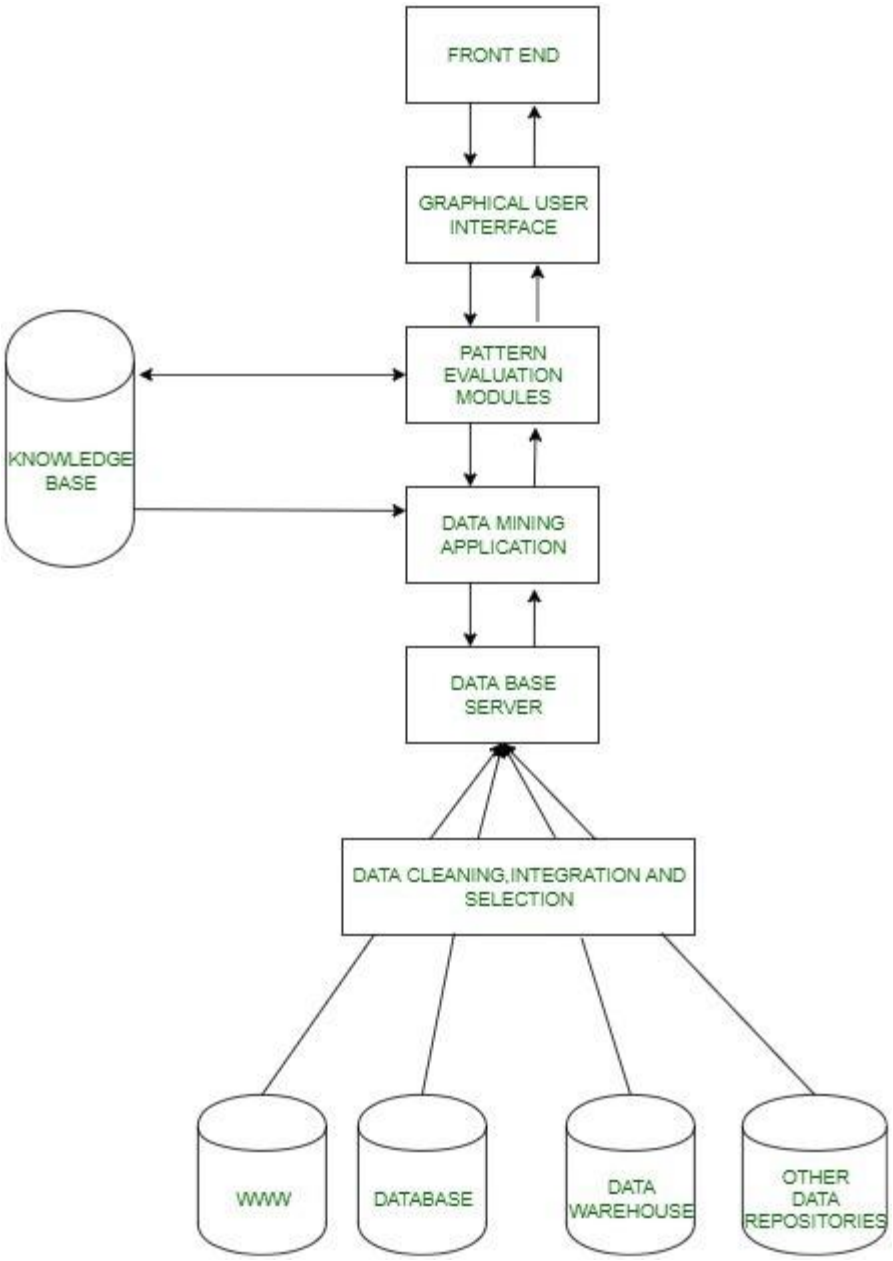
- It helps in exploring the large increase in the database and gather only valid information by improving segmentation.
- It's an efficient, cost-effective solution by uncovering the risk and fraud that makes profitable production.
- Sometimes customers having difficulty while purchasing helps in decision making and increases the sale.
- Data mining techniques can help organizations in real-time plan and save time.
- Also, saved money through fraud detection.

Application area of data mining:

- Future Healthcare
- Market Basket Analysis
- Manufacturing Engineering
- Fraud Detection
- Intrusion Detection
- Customer Segmentation
- Financial Banking

Data mining architecture:

Data mining architecture unveils how data extraction falls out. The architecture consists of various steps i.e Data source, data mining engine, data warehouse server, pattern evaluation, user interface, and knowledge base.



How data mining influences privacy, security, and socially:

Security and privacy have been an initial concern all the time. It aimed at future predictions using previous data. Suppose we buy any product so based on past purchases they make predictions and which also target our personal information. The continuous development of data mining techniques brings serious threats to data security and privacy which is very important to protect. The real threat is that if information gets exposed to unauthorized parties, it will be impractical to stop misuse. Therefore, we must need a system that possesses to protect data and its resources concerning authenticity and integrity.

How we can protect our data:

- Due to minimal protection setup, we lose data so we need to initiate a multilayer security system
- Access Controls are only given to those who have been given the authorization can access the data
- Data must verify an individual user's identity

So, some privacy preservation methods protect sensitive or private data while allowing useful information to be extracted from the data set.

- **Privacy-Preserving Data Mining (PPDM):** The main objective of the PPDM is to protect the privacy of the data and extract only relevant information. It ensures the protection of individual data to conserve privacy and provide accuracy by performing all the data mining operations.

Techniques of PPDM is further divided into various categories:

- **Data Hiding Technique:** In, this technique the data is reform in such a way that the sensitive or private information will not be visible to other parties. Using various ways we can implement these techniques such as Cryptographic Technique, Data Perturbation, and Anonymization Technique.
- **Knowledge hiding Technique:** In this technique, sensitive content is extracted from data using a data mining algorithm. There are different ways of implementing these techniques such as Association Rule Hiding, and Query Auditing.
- **Hybrid Technique:** It is a combination of the two techniques which infuse the limitations of the above two techniques.

SOCIAL IMPACTS OF DATA MINING

Data mining has innovatively influenced our daily lifestyle like how we work, shop, what we buy, search for any information, importantly saves our precious time and offers personalized product recommendations based on our previous history like amazon, Flipkart, etc.

Data mining emerging in all fields like Healthcare, Finance, Marketing, and social media. But there is a higher contribution towards healthcare and well-being by using data mining software to analyze data when developing drugs and to find associations between patients, drugs, and outcomes. And improving patient satisfaction, providing more patient-centered care, and decreasing costs, and increase operating efficiency and Insurance organizations can detect medical insurance fraud and abuse through data mining and reduce their losses.

An old payment system has now taken different forms of transactions depending on usage, acceptability, methods, technology, and availability. It changes the physical financial transactions to virtual payment transactions. So, data mining focuses on successful transactions and keeps track of fake transactions.

It is also used in Web-wide tracking technology that tracks user's interests while visiting any site. So, information about every site is been recorded, which can be used further to provide marketers with information reflecting your interests.

It is also used for customer relationship management which helps in providing more customized, personal service to individual customers. By studying browsing and purchasing history on Web stores, companies can tailor advertisements and promotions to customer profiles, only those who are interested and less likely to be annoyed with unwanted mailings. This helps in reducing costs, the waste of time, and improving work productivity.

Advantages:

Improved security: Data mining can help identify patterns and anomalies that could indicate security breaches, enabling organizations to take action to prevent future attacks.
Personalized services: Data mining can enable personalized services, such as customized product recommendations or personalized healthcare treatments, that can improve the overall customer experience.
Improved decision making: Data mining can provide insights into customer behavior, market trends, and other important factors, enabling organizations to make more informed decisions.
Improved efficiency: Data mining can help streamline business processes, enabling organizations to operate more efficiently.

Disadvantages:

Privacy concerns: Data mining can raise privacy concerns, as it involves collecting and analyzing data about individuals without their explicit consent. This can result in the disclosure of sensitive information, which can have negative consequences for individuals.
Security risks: The data used in data mining can be subject to security risks, such as unauthorized access or hacking, which can result in the exposure of sensitive information.
Bias: Data mining algorithms can be biased, which can lead to discriminatory outcomes or reinforce existing biases.
Social impacts: Data mining can have social impacts, such as increasing surveillance and reducing personal autonomy, which can have negative consequences for society as a whole.

What are the social implications of data mining?

Data mining is the process of finding useful new correlations, patterns, and trends by transferring through a high amount of data saved in repositories, using pattern recognition technologies including statistical and mathematical techniques. It is the analysis of factual datasets to discover unsuspected relationships and to summarize the records in novel methods that are both logical and helpful to the data owner.

Data mining systems are designed to promote the identification and classification of individuals into different groups or segments. From the aspect of the commercial firm, and possibly for the industry as a whole, it can interpret the use of data mining as a discriminatory technology in the rational search of profits.

There are various social implications of data mining which are as follows −

Privacy − It is a loaded issue. In current years privacy concerns have taken on a more important role in American society as merchants, insurance companies, and government agencies amass warehouses including personal records.

The concerns that people have over the group of this data will generally extend to some analytic capabilities used to the data. Users of data mining should start thinking about how their use of this technology will be impacted by legal problems associated with privacy.

Profiling − Data Mining and profiling is a developing field that attempts to organize, understand, analyze, reason, and use the explosion of data in this information age. The process contains using algorithms and experience to extract design or anomalies that are very complex, difficult, or time-consuming to recognize.

The founder of Microsoft's Exploration Team used complex data mining algorithms to solve an issue that had haunted astronomers for some years. The problem of reviewing, describing, and categorizing 2 billion sky objects recorded over 3 decades. The algorithm extracted the relevant design to allocate the sky objects like stars or galaxies. The algorithms were able to extract the feature that represented sky objects as stars or galaxies. This developing field of data mining and profiling has several frontiers where it can be used.

Unauthorized Used − Trends obtain through data mining designed to be used for marketing goals or some other ethical goals, can be misused. Unethical businesses or people can use the data obtained through data mining to take benefit of vulnerable people or discriminate against a specific group of people. Furthermore, the data mining technique is not 100 percent accurate; thus mistakes do appear which can have serious results.